NAS for VSR

Currently, we are actively investigating Neural Architecture Search (NAS) techniques tailored for lightweight Video Super-Resolution (VSR) on mobile platforms. Addressing the critical challenge of balancing real-time inference latency with high-fidelity reconstruction quality on resource-constrained devices, we have developed an automated architecture search framework. This framework is designed to adaptively identify optimal lightweight network topologies without manual intervention. Here are the preliminary experimental results:

Table 1: Comparison of different methods on PSNR, Runtime, and Score on REDS.

Methods	PSNR/dB (Val/Test)	Runtime/ms	Score (Val)
Bicubic	26.2986 / 26.50	-	-
NTV	- / 27.97	-	-
Rainbow	- / 27.99	18.0	-
Diggers	- / 28.33	19.9	-
NCUT VGroup	26.8135 / 27.46	10.2	0.0757
MVideoSR	26.9496 / 27.34	7.91	0.1179
ZX VIP	27.3058 / 27.52	8.39	0.1821
EVSRNet	27.4203 / 27.85	17.3	0.1035
EVSRNet w/o resize	27.4203 / 27.85	12.1	0.1480
EvoSR	27.6219 / -	10.8	0.2193
SYENet	26.7212 / -	8.08	0.0841
VPEG_R	26.8402 / -	11.8	0.0679
RESP	26.7804 / -	7.14	0.1033
RepTCN	27.0808 / -	8.14	0.1374
Our	**27.7794** / -	12.3	**0.2397**

Preliminary experimental results indicate that this approach significantly reduces computational delay while preserving superior super-resolution performance, demonstrating substantial potential for practical deployment. In future work, we aim to further enhance the efficiency of our search strategies and expand the diversity of the search space. Furthermore, we plan to generalize this automated framework to a broader range of resource-limited edge computing scenarios and related low-level vision tasks, thereby facilitating more extensive real-world applications. We welcome your attention to our future work.

Table 1: Comparison of Real-Time On-Mobile VSR Methods on Other Dataset in RGB (PSNR/SSIM)

Methods	REDS4	UDM10	Vimeo-90K-T
Bicubic	26.13 / 0.7295	30.91 / 0.8819	29.77 / 0.8482
NCUT VGroup	26.63 / 0.7514	31.65 / 0.8920	30.48 / 0.8607
MVideoSR	26.76 / 0.7553	31.81 / 0.8929	30.45 / 0.8608
ZX VIP	27.14 / 0.7694	32.33 / 0.9011	30.90 / 0.8685
EVSRNet	27.19 / 0.7705	32.39 / 0.9011	30.92 / 0.8672
SYENet	26.39 / 0.7441	31.26 / 0.8912	30.02 / 0.8584
VPEG_R	26.66 / 0.7534	31.82 / 0.8963	30.78 / 0.8682
RESR	26.58 / 0.7513	31.73 / 0.8952	30.60 / 0.8663
RepTCN	26.91 / 0.7643	32.17 / 0.9002	31.04 / 0.8932
EvoSR	27.49 / 0.7810	32.84 / 0.9093	**31.49 / 0.8793**
Our	**27.56 / 0.7831**	**32.91 / 0.9111**	31.47 / 0.8784